

Diffuse Optics - Restoring point source locations from surface radiance

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May 2024

Abstract

In this paper I present an inverse problem for finding illuminating point sources in a tissue from its surface radiance, by assuming that its optical properties are known. I am using the diffusion approximation and fitting the surface radiance to a superposition of green functions from several point sources.

1 Introduction

Finding the position of illuminating point sources inside a tissue is important to evaluate location of specific cells in a tissue. If surgeons would know where light is coming from, they will be able to extract only the harmful parts of tissue instead removing it completely. In previous research, a laser beam had been used to illuminate the tissue and find the tissue's optical properties by examining the radiance at the tissue surface[1, 2]. In this work, I try to do the opposite: by assuming that the optical properties are known, I try to find the location of the point sources that created this radiance. Assuming a uniform illumination in all of the tissue, I try to find the location of several fluorescent beads using well known diffusion green functions.

In this paper, I first present the problem mathematically, then, introduce the optimization algorithm and show how it works for a single point source. Then, I extrapolate the algorithm to work with multiple point sources. Afterwards, I present cases where the restoration does not succeed and finally examine the case where there is an estimation error in the

assumption of the tissue's optical properties.

2 Results

The setup for the problem is shown in Figure 1. Fluorescent point sources (beads) are illuminated uniformly. The camera is focused at the surface of the tissue and captures the radiance. I am using 2d setup to simplify the system, as the algorithm can be expanded to 3D.

2.1 Restoration of a single bead

In the presence of illuminating beads, it is common to use the diffusion approximation green function for the radiance.

$$\phi_i(r) = \frac{1}{4\pi D \cdot |r - r_i|} \cdot e^{-\mu_{eff} \cdot |r - r_i|} \quad (1)$$

where ϕ_i is radiance from bead i located at r_i , D is the diffusion coefficient and μ_{eff} is effective attenuation coefficient. The radiance at the tissue surface is:

$$\phi_i(x) = \frac{1}{4\pi D \cdot \sqrt{z_i^2 + (x - x_i)^2}} \cdot e^{-\mu_{eff} \cdot \sqrt{z_i^2 + (x - x_i)^2}} \quad (2)$$

where $r_i = (x_i, z_i)$, z_i is the depth of the bead and x_i its x axes position.

One can notice that for a single bead, its maximum is located at $x = x_i$ and its value is

$$\phi_i(x_i) = \frac{1}{4\pi D z_i} \cdot e^{-\mu_{eff} \cdot z_i} \quad (3)$$

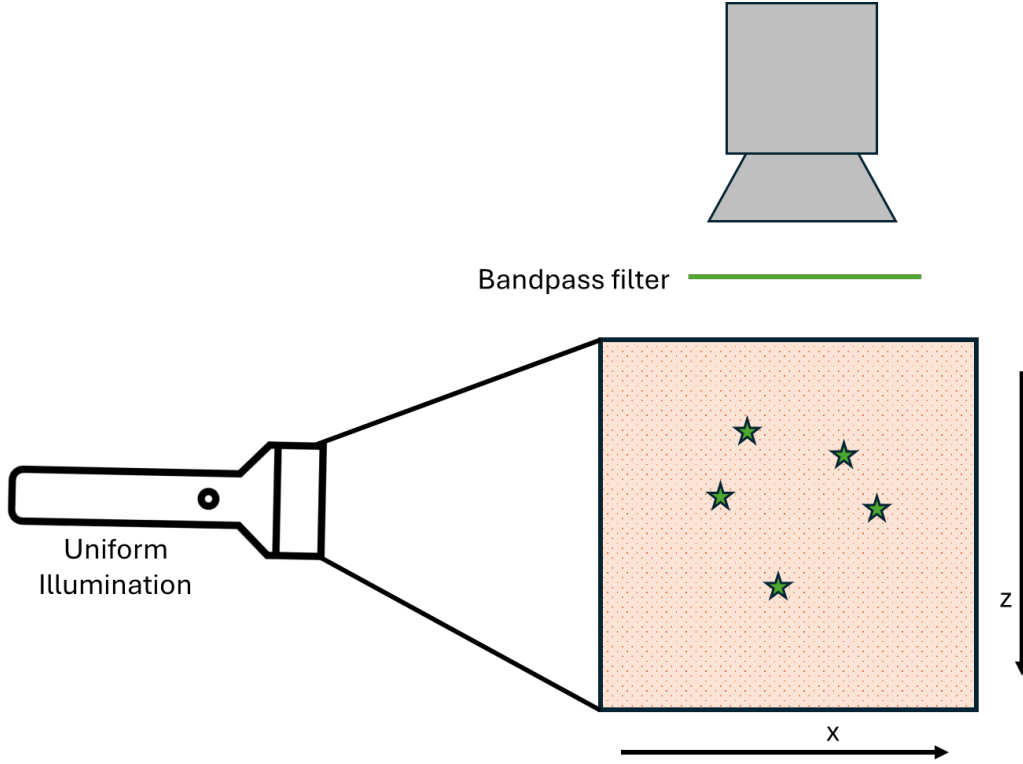


Figure 1: The setup is a 2d tissue with several fluorescent beads spread uniformly in the tissue. The tissue is illuminated with spatially incoherent light source, such that the radiance that reaches each bead is constant. The camera is focused at the tissue surface, and a band pass filter is placed such that it captures only the fluorescent light.

Assuming there is only one bead, finding x_i from a noiseless signal is relatively simple as it is the location of the maximum. To find z_i , I use Lambert-W-function, which solves the problem $ae^a = b$ for a . To reduce noise sensitivity, I use the vicinity of the signal's maximum to fit a parabola, and by using simple algebra find the parabola's maximum. This also allows the maximum to be between the samples, which is more accurate. Figure 2 provides a graphical demonstration for the above information, and Algorithm 1 provides a detailed explanation, where n_{vic} is the number of samples in the signal's vicinity.

Algorithm 1 Extracting x_i and z_i from surface radiance.

- 1: $m_x \leftarrow \arg \max_x \{\phi_i(x)\}$
 - 2: $y \leftarrow \phi_i[m_x - n_{vic} : m_x + n_{vic}]$
 - 3: $[a, b, c] \leftarrow \arg \min_{a,b,c} \{||ax^2 + bx + c - y||^2\}$
 - 4: $M \leftarrow \max\{ax^2 + bx + c\}$
 - 5: $W_{arg} \leftarrow \frac{\mu_{eff}}{4\pi D \cdot M}$
 - 6: $\hat{x}_i \leftarrow m_x + \frac{-b}{2a}$
 - 7: $\hat{z}_i \leftarrow LambertW(W_{arg}) \cdot \frac{1}{\mu_{eff}}$
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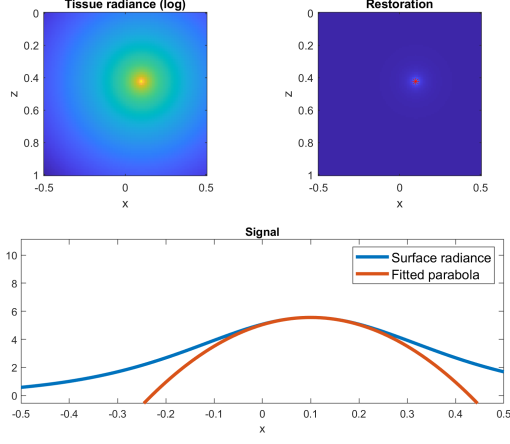


Figure 2: Restoration of a single point source location. The top-left image is the radiance (log scale) from the bead in the tissue. The top-right is the restoration, where a red asterisk is plotted at the estimated location. The bottom plot is the surface radiance as seen by the camera (blue), and the fitted parabola (orange).

2.2 Restoration of a multiple beads

The case where more than a single bead exists in the tissue is examined in this subsection. To find multiple beads, the strategy used here is such: assume that the signal is made only of a single bead and try to fit its location. Then, model the surface radiance from this estimated bead and subtract the modeled signal from the actual signal. If the restoration is accurate, it should contain now the surface radiance from the other sources. Repeat this process until the signal equals zero. Figure 3 provides an example for a successful restoration of 3 beads and Algorithm 2 provides a detailed explanation.

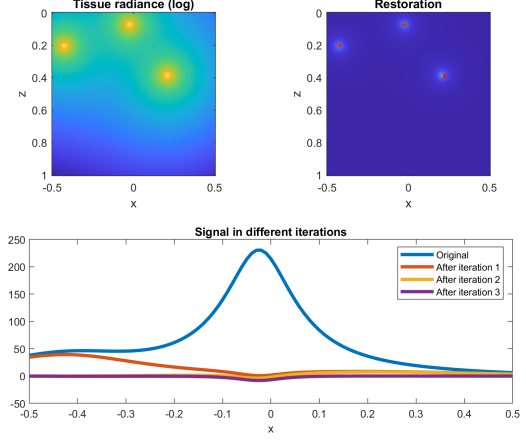


Figure 3: Restoration of 3 beads locations. The top-left image is the radiance (log scale) of the beads in the tissue. The top-right is the restoration, where red asterisks are plotted at the estimated locations. The bottom plot is the signal in some iterations.

Algorithm 2 Estimating the location of multiple beads.

- 1: $\hat{I} \leftarrow$ Tissue surface radiance
 - 2: $i = 1$
 - 3: $[M, z_1, x_1] \leftarrow$ Single bead approximation from signal $\hat{I}$
 - 4: **while** $M > threshold$ **do**
 - 5: $R_i = \frac{1}{4\pi D \cdot \sqrt{z_i^2 + (x - x_i)^2}} \cdot e^{-\mu_{eff} \cdot \sqrt{z_i^2 + (x - x_i)^2}}$
 - 6: $\hat{I} \leftarrow \hat{I} - R_i$
 - 7: $i = i + 1$
 - 8: $[M, z_i, x_i] \leftarrow$ Single bead approximation from signal $\hat{I}$
 - 9: **end while**
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There is an issue with this method. Since the signal is not really from a single bead, the parabola approximation is not accurate and there is a slight error in the restoration. Hence, the modeled signal is slightly shifted in x and z than the actual signal, such that af-

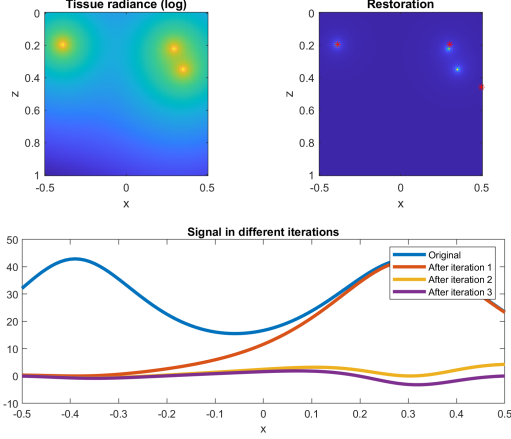


Figure 4: Restoration of 3 beads locations. The top-left image is the radiance (log scale) of the beads in the tissue. The top-right is the restoration, where red asterisks are plotted at the estimated locations. The bottom plot is the signal in some iterations. Notice that there are two beads approximately close in the x axes to the right. The algorithm finds correctly the x location of the upper bead, but its z location is slightly wrong. This is because the signal from the lower bead is added to it and it is no longer valid for a single bead approximation. The restoration for the lower bead is far worse, since its signal is completely hindered in the error of the upper bead.

ter the subtraction, a portion of the signal from this bead still exists. Since the signal is decaying very quickly in z , signal from deeper parts of the tissue is much weaker and can be hindered in the restoration error of the stronger bead. Figure 4 shows such an example.

2.3 Sensitivity to the optical properties assumption errors

In this paper I have assumed to know the optical properties of the tissue. In this section, the sensitivity of the restoration to errors in them is analyzed. The diffusion approximation model uses $\mu_{eff} \approx \sqrt{\mu_a \mu'_s}$

and $D = \frac{1}{3 \cdot (\mu_a + \mu'_s)}$. To find the bead x location, only the maximum of the signal is used, which is not affected by these parameters. For estimation error in μ_s , such that $\hat{\mu}_s = \epsilon_s \mu_s$, the expression for $\hat{z}$ is

$$\hat{z} = \frac{L\left(\frac{\sqrt{\mu_a \cdot \hat{\mu}'_s} \cdot 3(\mu_a + \hat{\mu}'_s)}{4\pi M}\right)}{\sqrt{\mu_a \cdot \hat{\mu}'_s}} \approx \frac{L(\epsilon_s \sqrt{\epsilon_s} \frac{\sqrt{\mu_a \cdot \mu'_s} \cdot 3(\mu_a + \mu'_s)}{4\pi M})}{\sqrt{\epsilon_s} \sqrt{\mu_a \cdot \mu'_s}} \quad (4)$$

Where $L(W)$ is the Lambert W function and M is the signal maximum. The argument inserted to the Lambert W function is between 0.05 for shallow beads, and around 7 for deep ones (in my setup). The derivative of the Lambert W function with argument of 0.05 is around 1, thus, an estimation error of ϵ_s in μ_s will cause the Lambert W function to deviate in $\epsilon_s \cdot \sqrt{\epsilon_s}$. It can be seen from the right expression in equation (4) that eventually, an error of ϵ_s will take place in the estimation of z . For deeper beads, the derivative of the Lambert W function is very small, causing a very small deviation with respect to changes in the argument. In this case, the error is around $\frac{1}{\sqrt{\epsilon_s}}$.

For estimation error in μ_a , such that $\hat{\mu}_a = \epsilon_a \mu_a$, the expression for $\hat{z}$ is:

$$\hat{z} = \frac{L\left(\frac{\sqrt{\hat{\mu}_a \cdot \mu'_s} \cdot (\hat{\mu}_a + \mu'_s)}{4\pi M}\right)}{\sqrt{\hat{\mu}_a \cdot \mu'_s}} \approx \frac{L(\sqrt{\epsilon_a} \frac{\sqrt{\mu_a \cdot \mu'_s} \cdot (\mu_a + \mu'_s)}{4\pi M})}{\sqrt{\epsilon_a} \sqrt{\mu_a \cdot \mu'_s}} \quad (5)$$

In this case, if the bead is close to the surface, error in μ_a will almost not affect the restoration, since the error in both the nominator and denominator. On the other hand, if the bead is deep in the tissue, the estimation error will be around $\frac{1}{\sqrt{\epsilon_a}}$.

3 discussion

In this work I presented a method to estimate location of fluorescent point sources from the tissue's surface radiance. While it works nicely for a single bead, in the presence of multiple beads the algorithm is not robust. It fails when there are too many beads, or even when the beads are hindered behind each other. However, it usually succeeds in finding

the approximate location of the beads. While I could have used correlations instead of parabola fitting, I found the later to be easier since the parabolas maximum is found with simple algebra, and allows it to be between samples.

According to the last subsection, the sensitivity of the algorithm to optical properties assumption errors is descent, as the estimation error is only in z , and its magnitude is around the same as the assumption errors. A MATLAB code is attached which simulates this setup and the algorithm.

References

- [1] S.-P. Lin, L. Wang, S. L. Jacques, and F. K. Tit-tel. Measurement of tissue optical properties by the use of oblique-incidence optical fiber reflectometry. *Applied optics*, 36(1):136–143, 1997.
- [2] G. Marquez, L. V. Wang, S.-P. Lin, J. A. Schwartz, and S. L. Thomsen. Anisotropy in the absorption and scattering spectra of chicken breast tissue. *Appl. Opt.*, 37(4):798–804, Feb 1998.